# Andrei Stefan Bejgu (he/him)

**Date of Birth:** 16-11-1995
**Location:** Rome, Italy
**Languages:** English (Fluent), Italian (Native), Romanian (Native)

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AI engineer and researcher with **9+ years of experience**. Today I am Senior AI Applied Scientist at Syllotips, working on how its agents plan, use tools, cite sources and improve over time. Before that I spent 8 years at Babelscape, growing from software engineer to NLP researcher, earned an industrial Ph.D. in AI at Sapienza and published at ACL, EMNLP, EACL and NAACL.

## Experience & Selected Projects

### Syllotips
*Remote*

**Senior AI Applied Scientist**
Oct. 2025–Present

- Redesigned and rebuilt the **deep-research agent**, the main revenue product: tool calling instead of fixed branches, shared working memory, and a harness with tool budgets and bounded replanning.
- Designed the **MCP** layer that connects the agents to enterprise tools such as Atlassian, HubSpot and Intercom: one architecture for cloud and private cloud, OAuth handling, and tool results the agents can cite.
- Built a sandboxed **code interpreter** that lets agents analyse spreadsheets and produce reports and files, on cloud and private cloud alike.
- Rebuilt the **assistant agent** that helps experts close the loop on issues: it investigates what went wrong and proposes a solution for them to review.
- Built **self-hosted model serving** on vLLM across multi-GPU clusters: open-weight LLMs with up to 1M-token context, plus small decision models the agents use for fast routing and grading, so the agents can run fully on our own infrastructure.

### Babelscape
*Rome, Italy*

**NLP Researcher**
Oct. 2021–Sep. 2025

I led advanced NLP research translating state-of-the-art insights into production AI products, with a focus on retrieval-augmented systems and multilingual LLM fine-tuning.

- Designed and developed real-time retrieval pipelines, maximizing precision and efficiency for Vera, a multilingual fact-checking system.
- **Distributed** (Slurm) fine-tuning of open-source LLMs for domain-specific tasks such as **text2sql and claim-extraction/verification**.
- Contributed to the supervised-fine-tuning stage of **Minerva**, the first Italian LLM.

**NLP Engineer**
Jun. 2020–Oct. 2021

- Designed and deployed Semantic Paths, a **Multilingual Semantic Search** engine for scientific articles written in Kotlin.
- Developed advanced **Word Sense Disambiguation** models for large-scale multilingual processing.

**Software Engineer**
Jul. 2017–Jun. 2020

- Developed myKnowledgeGraph, a **Knowledge Graph Extraction** and visualization system for enterprise data.
- Designed the company's **Java NLP Pipeline**, maximizing speed, accuracy, and scalability of multilingual text processing.

### CINI - Consorzio Interuniversitario Nazionale per l'Informatica
*Rome, Italy*

**Software Developer**
Dec. 2016–Jan. 2017

Created an advanced tool for lexicon-semantic stylometric analysis, enabling efficient processing of large-scale texts.

## Education

**Ph.D. in Artificial Intelligence**
*Sapienza University of Rome* Jan. 2025
**Focus:** Explored multilingual NLP by applying lexical semantics to modern systems. Improved domain-specific tasks, looked into complex linguistic patterns, and worked on word- and sentence-level representations in RAG systems and LLMs. This was a joint industrial Ph.D., combining academic research with industry-driven projects. Created context-aware, domain-specific embeddings that boosted retrieval performance. –**Advisor:** Prof. R. Navigli

**MSc in Computer Science**
*Sapienza University of Rome –**Grade**: 110/110 with honors* Nov. 2021
**Thesis:** Exploring language-agnostic latent spaces for sentences. –**Advisor:** Prof. R. Navigli

**BSc in Computer Science**
*Sapienza University of Rome –**Grade**: 107/110* Jun. 2019
**Thesis:** Knowledge-Based Word Sense Disambiguation leveraging synset embeddings. –**Advisor:** Prof. R. Navigli

## Publications

[1] A. Scirè*, **A. S. Bejgu***, S. Tedeschi, K. Ghonim, F. Martelli, and R. Navigli. "Truth or Mirage? Towards End-to-End Factuality Evaluation with LLM-OASIS". In: *Computational Linguistics* (2025).

[2] L. Moroni, T. Bonomo, L. Gioffré, L. Xu, D. Fedele, L. Colosi, **A. S. Bejgu**, A. Scirè, and R. Navigli. "What we Learned from Continually Training Minerva: a Case Study on Italian". In: *CLiC-it*. 2025.

[3] L. Moroni, G. Puccetti, P.-L. H. Cabot, **A. S. Bejgu**, A. Miaschi, E. Barba, F. dell'Orletta, A. Esuli, and R. Navigli. "Optimizing LLMs for Italian: Reducing Token Fertility and Enhancing Efficiency Through Vocabulary Adaptation". In: *Findings of NAACL*. 2025.

[4] K. Ghonim, **A. S. Bejgu**, and R. Navigli. "Conceptpedia: Leveraging Traditional Approaches for Multimodal Image-Text Representation". In: *EMNLP*. 2025.

[5] **A. S. Bejgu**, E. Barba, L. Procopio, A. Fernández-Castro, and R. Navigli. "Word Sense Linking: Disambiguating Outside the Sandbox". In: *ACL*. 2024.

[6] F. Molfese, **A. S. Bejgu**, S. Tedeschi, S. Conia, and R. Navigli. "CroCoAlign: A Cross-Lingual, Context-Aware and Fully-Neural Sentence Alignment System for Long Texts". In: *EACL*. 2024.

[7] F. Martelli, **A. S. Bejgu**, et al. "XL-WA: a Gold Evaluation Benchmark for Word Alignment in 14 Language Pairs". In: *CLiC-it*. 2023.

## Other Activities

**Reviewer Activity**: Reviewer for EACL (2023), ACL (2023, external reviewer), EMNLP (2023, external reviewer), ICLR (2024, external reviewer), and ACL ARR (2024 and 2025).
**Teaching Activity at Sapienza University of Rome**:

- **Natural Language Processing Course (2022–2023)**: Designed and led practical lab sessions and assignments, guiding master's students through state-of-the-art NLP techniques.
- **Hot Topics in NLP (2023–2024)**: Developed and managed one of the three major course assignments, providing hands-on exposure to cutting-edge advancements in NLP.

## Scholarships and Awards

- Best Employee Award, Babelscape 2021
- Merit Scholarship Winner, LazioDiSCo.it 2018–2020